Software Testing is the process of evaluating a software application to find defects, verify requirements, and ensure it works as expected before release. It involves executing the system (manually or via automation) and checking whether actual behavior matches expected behavior.

You can think of it as: *"checking software quality by validating and breaking it before users do."*

Why software testing is necessary

- **Finds defects early** → reduces cost of fixing bugs after release
- **Ensures requirements are met** → builds the right product, not just a working one
- **Improves product quality** → stability, performance, usability
- **Prevents business loss** → avoids failures in production (money, reputation, users)
- **Enhances security** → detects vulnerabilities before attackers do
- **Builds user trust** → reliable software increases adoption

Quality and Quality Assurance

- **Quality**: A degree of excellence of a product or a service.
- **Quality assurance (QA)**: is any systematic process of determining whether a product or service meets specified requirements.